Comprehensive Research Analysis: IIT Bombay Department of Electrical Engineering - Current and Recent Research Activities

Based on extensive analysis of faculty profiles, recent publications, and current research projects spanning 2020-2024, the Department of Electrical Engineering at IIT Bombay represents one of India's most comprehensive and cutting-edge electrical engineering research environments, encompassing theoretical and experimental work across multiple frontiers of modern electrical engineering.

Advanced Control Systems and Multidimensional Theory

Prof. Harish K. Pillai (Head of Department) leads groundbreaking research in multidimensional control systems, behavioral control theory, and coding theory applications. His recent revolutionary work focuses on analysis of atypical orbits in one-dimensional piecewise-linear discontinuous maps, third-order bimodal switched systems, and comparison between different notions of stability for Laurent systems. His cutting-edge contributions include behavior of infinite chains of kinematic points with immediate-neighbors interaction dynamics, FVTD large time-step method using Radon Transform for electromagnetic field computations, and permutation polynomial representatives with their matrices. Prof. Pillai's research spans stability analysis of discrete 2-D autonomous systems, multidimensional systems theory, numerical and computational methods, optimization techniques, and advanced electromagnetics.[247][248][249]

Prof. Madhu N. Belur specializes in dissipative dynamical systems, structural controllability in heterogeneous subsystems, and graph-theoretic control methods. His recent breakthrough research includes sparsest feedback selection algorithms for structurally cyclic systems with dedicated actuators and sensors, algebraic connectivity optimization for local and global maximizer graphs, and rationalized timetabling using simulation tools representing a paradigm shift in Indian Railways. His group's cutting-edge work encompasses optimal network topology design in composite systems for structural controllability, interlacing properties of system-poles and system-zeros in MIMO systems, and development of railway timetabling and capacity estimation tools for Indian Railways' Zero Base Time-Tabling initiative.[250][251][247]

Prof. Debraj Chakraborty focuses on optimal control theory, differential games, and nuclear reactor control systems. His innovative research includes time-optimal feedback control for pursuit-evasion games, intermittent feedback control for disturbance rejection, optimized structured state feedback controllers for nuclear power reactors including bulk power control of large-core PHWRs and zone power control systems. His recent breakthrough work covers learning from similar systems using online data-driven LQR with iterative randomized data compression, randomized subspace identification for LTI systems, and random forests for time-optimal feedback control in pursuit evasion games.[252][247]

Power Electronics and Renewable Energy Systems

Prof. Vivek Agarwal leads cutting-edge research in photovoltaic power conversion systems, particularly under partially shaded conditions. His recent groundbreaking work includes improved dual boost ANPC-type five-level inverter topologies, optimal distributed generation planning incorporating energy management for economical and resilient smart distribution systems, design of robust PID controllers using PSO-based automated QFT for non-minimum phase boost converters, and novel leg-integrated switched capacitor inverter topologies for three-phase induction motor drives. His research spans power quality issues, non-conventional energy systems including solar PV, wind, and fuel cells, with applications in maximum power point tracking, stand-alone and grid-connected systems, and microgrids.[253][254][247]

Prof. Sandeep Anand specializes in wide bandgap semiconductor applications, focusing on normally-on GaN and normally-on SiC MOSFET gate driving circuits. His revolutionary contributions include ultrafast self-powered circuits for gate driving of normally-on wide bandgap transistors, voltage regulation and load sharing in DC microgrids using single variable global average estimation, gate voltage-based active thermal control of power semiconductor devices, and impact of operational parameters on dVDS/dt of SiC MOSFETs with schemes for gate driver resistance selection. His cutting-edge research encompasses reliability of power electronic circuits, circuits for interfacing alternate energy sources, electric vehicle drive trains and chargers, and development of highly efficient GaN-based PV inverters with reduced leakage current.[255][256][247]

Prof. Shiladri Chakraborty works on high-frequency isolated converters, soft-switching techniques, and multi-physics converter design optimization. His recent breakthrough research includes comparison of CCM and CRM-based boost parallel active power decouplers for PV microinverters, characterization of bare-die SiC-based, wirebond-less, integrated half-bridge with multi-functional bus-bar, DAB converters for EV on-board chargers using bare-die SiC MOSFETs and leakage-integrated planar transformers. His innovative work encompasses high frequency link ripple power compensation strategies for single-phase bidirectional AC-DC matrix converters, design optimization for weighted conduction loss minimization in dual-active-bridge-based PV microinverters, and development of wide bandgap devices including GaN and SiC-based power converters.[257][258][259][247]

Prof. Baylon G. Fernandes focuses on advanced inverter topologies, permanent magnet machines for wind power, and switched reluctance machines for electric vehicles. His recent work includes estimation of resonance frequencies using electrostatic energy-based capacitance models for medium/high-frequency transformers, evaluation of multi-frequency power electronic converters including concept, architectures and realization, and effect of material resistivity and temperature on leakage inductance of medium frequency transformers made of aluminum and copper foils.[247]

Prof. Himanshu J. Bahirat specializes in offshore wind energy systems, HVDC transmission networks, and DC wind farms. His recent research covers robust decoupled outer-control design for single VSC and HVDC grids with weak AC system interconnection, analytical approach for optimal HTS solenoid design, multi-terminal DC networks, and transient analysis in renewable energy integration. His work includes series-connected offshore wind turbines, DC power systems reliability assessment, and controller interaction and stability margins in mixed SCR MMC-based HVDC grids.[260][261][247]

Communication Systems and Signal Processing

Prof. Kumar Appaiah focuses on MIMO communication systems, fiber optic communications, and advanced signal processing. His cutting-edge work includes GSEIM general-purpose simulator with explicit and implicit methods, low complexity FBMC for wireless MIMO systems, post-coder feedback approaches for MIMO-TDD training asymmetry, and surface engineering using titanium nanocoatings for silicon optical devices. His research encompasses signal processing for optical communication, multiplexing in wireless and fiber-optic communication systems, and predictive quantization and joint time-frequency interpolation techniques for MIMO-OFDM precoding.[262][263][247]

Prof. Siddharth Tallur conducts cutting-edge research in embedded systems and sensors, digital systems and signal processing, photonics and frequency control, and microsystems and MEMS. His current projects include electrochemical sensors for DNA and protein sensing, AMR systems, water quality monitoring, acoustic and electromagnetic techniques for corrosion monitoring, and ultrasonic guided wave structural health monitoring. His recent breakthrough work encompasses electrochemical sensing of SARS-CoV-2 amplicons with PCB electrodes, TinyML-enabled edge implementation of transfer learning framework for domain generalization in machine fault diagnosis, unsupervised learning framework for temperature compensated damage identification and localization in ultrasonic guided wave SHM, and performance evaluation of dipstick-based wastewater surveillance of viruses.[264][265][247]

Prof. Rajbabu Velmurugan specializes in statistical and digital signal processing, audio and speech processing, image processing and computer vision. His recent work includes all-pass filter design using unimodular interpolation, multi-target unsupervised domain adaptation for person re-identification, and scalable implementation of particle filter-based visual object tracking on network-on-chip systems.[247]

Prof. Saravanan Vijayakumaran focuses on signal processing for communications and parallel simulation algorithms. His innovative research includes MProve+ privacy enhancing proof of reserves protocol for Monero, properties of syndrome distribution for blind reconstruction of cyclic codes, and GOSPAL efficient strategy-proof mechanism for constrained resource allocation.[247]

VLSI Design and Emerging Technologies

Prof. Maryam Shojaei Baghini leads research in analog/mixed-signal VLSI design with applications in AI/ML, neuromorphic systems, and healthcare. Her recent groundbreaking contributions include current-excitation based delta-R to frequency converters for resistive sensors, band-to-band tunneling-based oscillator-enabled temperature-to-digital converters with resolution FoM of 0.16 pJK^2 for embedded temperature sensing, wearable accelerometer-based systems for knee angle monitoring during physiotherapy, and gait phase identification using wearable sensors for physiotherapy assistance. Her cutting-edge research encompasses AI/ML domain applications, circuits and systems for neuromorphic applications, low voltage/low power/low energy designs for healthcare, bio-inspired circuits and systems, instrumentation, energy harvesting, and high-frequency integrated circuit design.[266][267][247]

Prof. Udayan Ganguly specializes in emerging memory technologies, neuromorphic computing architectures, and transistor variability analysis. His research includes PCMO-based RRAM for integrate-and-fire neurons in spiking neural networks, stochastic neurons for Boltzmann machines to solve maximum cut problems, transient Joule heating-based oscillator neurons for neuromorphic computing, and analytical modeling of metal gate granularity-induced variability in nanowire FETs.[247]

Prof. Debanjan Bhowmik focuses on neuromorphic computing using spintronic devices, non-volatile memory technologies, and quantum computing applications. His cutting-edge research includes impact of edge defects on synaptic characteristics of ferromagnetic domain-wall devices for on-chip learning, oscillator-synchronization-based off-line learning algorithms with on-chip inference using spin Hall nano-oscillator arrays, demonstration of synaptic behavior in heavy-metal-ferromagnetic-metal-oxide-heterostructure-based spintronic devices for crossbar-array-based neural networks. His group works on nanomagnetism and spintronics, quantum computing for artificial intelligence, and development of brain-inspired hardware for energy-efficient AI applications.[268][269][270][247]

Optoelectronics and Quantum Devices

Prof. Subhananda Chakrabarti leads III-V compound semiconductor research, quantum dot photodetectors, and solar cells. His groundbreaking work includes tuning structural and optical characteristics of heterogeneously coupled InAs submonolayer quantum dots grown on InAs Stranski-Krastanov quantum dots, comprehensive analysis of strain profile in heterogeneously coupled Stranski-Krastanov on submonolayer quantum dot heterostructures, study on inter-band and inter-sub-band optical transitions with varying InAs/InGaAs sub-monolayer quantum dot heterostructure stacks, and investigation of inter-dot tunneling effects in hybrid coupled quantum dots for enhancing performance. His research encompasses III-V device integration on germanium, molecular beam epitaxy growth techniques, and establishment of India's first high-temperature operating GaAs-based thermal imaging devices responding at long wavelengths capable of detecting human objects.[271][272][273][274][247]

Advanced Signal Processing and Pattern Recognition

Prof. Subhasis Chaudhuri works on computer vision, haptic rendering, and biomedical signal processing. His recent research includes haptic rendering of oriented point clouds of heritage objects using proxy projection, scene recognition from optical remote sensing using deep feature mining, graph convolutional networks for multi-label VHR remote sensing, and QoS-compliant telehaptic communication over shared networks.[247]

Information Theory and Network Systems

Prof. Bikash Kumar Dey specializes in information theory, coding theory, and secure communications. His work covers communication in the presence of state-aware adversaries, wiretapped oblivious transfer protocols, zero-error function computation through bidirectional relays, and oblivious transfer over wireless channels.[247]

Prof. Prasanna Chaporkar focuses on resource allocation and scheduling in wired/wireless networks, optimization and control of stochastic systems. His recent research includes sparsest feedback selection for structurally cyclic systems, approximating constrained minimum cost input-output selection for generic arbitrary pole placement in structured systems, and GOSPAL efficient strategy-proof mechanism for constrained resource allocation.[247]